Huadai Liu

Multimodal Foundation Models • Audio Language Models • Post-training

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SUMMARY

Researcher working on multimodal foundation models that perceive, reason, and generate: audio language models for speech, sound, and music; multimodal LLM reasoning; post-training with reinforcement learning, on-policy distillation, and chain-of-thought; and the latent representations and few-step samplers beneath diffusion and flow-based generation. 16 first-author papers at NeurIPS, ICML, ICLR, ACL, EMNLP, and ACM MM.

RESEARCH HIGHLIGHTS

Audio Language Models

- **AudioCALM** (NeurIPS 2026). One continuous autoregressive model for speech, sound, and music: a flow-matching head replaces the softmax and block-causal attention supports streaming; beats CosyVoice 3.0 on WER (2.0% vs. 2.2%) and TangoFlux on FAD (1.95 vs. 2.70).

Multimodal LLMs: Reasoning & Evaluation

- **ThinkSound** (NeurIPS 2025). Multimodal-LLM chain-of-thought binds visual events to acoustic conditions before flow-matching synthesis, and drives object-level refinement and instruction-based editing; FD 34.6 vs. 43.3 for MMAudio on VGGSound, with 2.8× faster inference.
- **X-Stream** (ECCV 2026, co-first author). First benchmark for multi-stream streaming understanding (4,220 QA pairs over 932 videos, 11 subtasks), analysing MLLMs as signal multiplexers; state-of-the-art MLLMs score about 50%.

Post-training & Alignment

- **PrismAudio** (ICLR 2026). Four reasoning modules with matching rewards; Fast-GRPO's hybrid ODE–SDE sampling exceeds Flow-GRPO's final reward in 200 steps (vs. 600+); best on all four perceptual dimensions.
- **MosaicAudio** (EMNLP 2026). Distills four flow-matching teachers into one controllable student with per-axis on-policy supervision and a perceptual manifold anchor.

Latent Representations & Fast Sampling

- **STAR-VAE** (ICML 2026). Structure–texture-aligned latent channels with a CNN–Mamba backbone beat the Stable Audio Open VAE on all reconstruction metrics at equal latent rate; text-to-audio FD 89.2 → 72.5.
- **FlashAudio / AudioLCM** (ACL 2025 Oral / ACM MM 2024). **Single-step generation at 400× real time**, beating diffusion baselines that use hundreds of steps; AudioLCM synthesizes in 2 steps at 333× real time.

OPEN SOURCE & IMPACT

- **CosyVoice 2** (core contributor). LLM-based multilingual streaming TTS; designed the rectified-flow inference path, a 5–10× inference speed-up in production; **23.9K GitHub stars**, **480+ citations**.
- **ThinkSound & PrismAudio** (research and project lead). Released models, AudioCoT data, the AudioCanvas benchmark, and full training code; **1.3K+ GitHub stars**, **1M+ social-media views**, **#3 on Papers with Code Trending**; covered by QbitAI and IT Home.
- **AudioLCM** (lead). Few-step text-to-audio training and inference code; **1.1K+ GitHub stars**.

EDUCATION

Zhejiang University M.S. in Artificial Intelligence — 2021 – 2024
Advisor: Prof. Zhou Zhao.

Shandong University B.S. in Electronic Information — 2016 – 2020

PUBLICATIONS

26 published or accepted papers; 16 first-author papers, including 4 co-first-author papers.
Google Scholar: 1,446 citations; h-index 16.

First & Co-first Author * equal contribution

1. **H. Liu**, K. Luo, W. Wang, et al. AudioCALM: Continuous Autoregressive Language Modeling for Universal Audio Generation. **NeurIPS 2026**.
2. **H. Liu**, W. Wang, K. Luo, et al. STAR-VAE: Structured Topology-Aware Regularization for Audio Reconstruction and Generation. **ICML 2026**.
3. **H. Liu**, K. Luo, W. Wang, et al. PrismAudio: Decomposed Chain-of-Thoughts and Multi-dimensional Rewards for Video-to-Audio Generation. **ICLR 2026**.
4. **H. Liu**, P. Sun, Q. Chen, et al. MosaicAudio: Multi-Teacher On-Policy Distillation for Controllable Video-to-Audio Generation. **EMNLP 2026**.
5. P. Sun*, X. Lu*, **H. Liu***, et al. X-Stream: Exploring MLLMs as Multiplexers for Multi-Stream Understanding. **ECCV 2026**.
6. **H. Liu**, K. Luo, J. Wang, et al. ThinkSound: Chain-of-Thought Reasoning in Multimodal Large Language Models for Audio Generation and Editing. **NeurIPS 2025**.
7. **H. Liu**, T. Luo, K. Luo, et al. OmniAudio: Generating Spatial Audio from 360-Degree Video. **ICML 2025**.
8. **H. Liu**, J. Wang, R. Huang, et al. FlashAudio: Rectified Flow for Fast and High-Fidelity Text-to-Audio Generation. **ACL 2025, Oral; SAC Highlight**.
9. **H. Liu**, J. Wang, X. Li, et al. MEDIC: Zero-shot Music Editing with Disentangled Inversion Control. **ACM MM 2025, Oral**.
10. **H. Liu**, R. Huang, Y. Liu, et al. AudioLCM: Efficient and High-Quality Text-to-Audio Generation with Minimal Inference Steps. **ACM MM 2024**.
11. **H. Liu**, R. Huang, J. He, et al. Wav2SQL: Direct Generalizable Speech-To-SQL Parsing. **Findings of ACL 2024**.
12. **H. Liu**, W. Xu, X. Lin, et al. AntCritic: Argument Mining for Free-Form and Visually-Rich Financial Comments. **LREC-COLING 2024**.
13. R. Huang*, J. Liu*, **H. Liu***, et al. TranSpeech: Speech-to-Speech Translation With Bilateral Perturbation. **ICLR 2023**.
14. **H. Liu**, R. Huang, X. Lin, et al. ViT-TTS: Visual Text-to-Speech with Scalable Diffusion Transformer. **EMNLP 2023**.
15. R. Huang*, **H. Liu***, X. Cheng, et al. AV-TranSpeech: Audio-Visual Robust Speech-to-Speech Translation. **ACL 2023**.
16. R. Huang*, Z. Zhao, **H. Liu***, et al. ProDiff: Progressive Fast Diffusion Model for High-Quality Text-to-Speech. **ACM MM 2022**.

Selected Co-authored

17. P. Sun, S. Lang, D. Wu, . . . , **H. Liu**, et al. SpaceVista: All-Scale Visual Spatial Reasoning from mm to km. **ICML 2026**.
18. F. Zhang, Z. Cheng, C. Deng, . . . , **H. Liu**, et al. MME-Emotion: A Holistic Evaluation Benchmark for Emotional Intelligence in Multimodal Large Language Models. **ICLR 2026**.
19. P. Sun, S. Cheng, X. Li, Z. Ye, **H. Liu**, et al. Both Ears Wide Open: Towards Language-Driven Spatial Audio Generation. **ICLR 2025**.
20. Z. Du, Y. Wang, Q. Chen, . . . , **H. Liu**, et al. CosyVoice 2: Scalable Streaming Speech Synthesis with Large Language Models. **Technical report, 2024**.

TECHNICAL SKILLS

Methods: Audio language models; multimodal LLMs; RL post-training and distillation; diffusion and flow matching; VAE and latent representation design.

Systems: PyTorch, FSDP, DeepSpeed, Ray; multi-node distributed training and streaming inference.

Programming: Python, C++.

HONORS & SERVICE

- **ACL 2025 Senior Area Chair Highlight** and Oral (FlashAudio); Oral at ACM MM 2025 (MEDIC); **ECCV 2026 Outstanding Reviewer**; National Scholarship (2023); Outstanding Graduate of Zhejiang Province (2024).
- **Area Chair**, ACL Rolling Review (2026). **Reviewer** for NeurIPS, ICML, ICLR, ACL ARR, EMNLP, ECCV, ACM MM, AAAI, and IEEE/ACM TASLP.